

Ideological Expansion and Paradox

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Introduction

The Algebra of Propositions

We assume propositions form a Boolean algebra $\mathcal{A}$:

- There is a greatest proposition $\top$ and a least proposition $\perp$.
- Each proposition p has a complement $\sim p$.
- Two propositions p and q have a join $p \cup q$ and a meet $p \cap q$.
- Join and meet distribute over each other.

$\mathcal{A}$ is a Boolean algebra $(\mathcal{A}, \cup, \cap, \sim, \top, \perp)$.

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Quantified Propositional Logic

The Language of QPL

Let $\mathcal{L}$ be the language of quantified propositional logic:

	expressions	axioms
variables	$p, q, r, \dots$	
constants	$\perp, \top$	
connectives	$\neg, \wedge, \vee$	classical tautologies
quantifiers	$\forall, \exists$	elementary quantification theory
identity	$=$	reflexivity, Leibniz's Law

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A Boolean-Valued Semantics for QPL

Given a Boolean algebra $\mathcal{A}$:

$$(A, \cap, \cup, \sim, \top, \perp).$$

$$x \leq y := x \cap y = x$$

An assignment α maps propositional variables into elements of A :

$\llbracket p \rrbracket_\alpha$	$:=$	$\alpha(p)$	
$\llbracket \top \rrbracket_\alpha$	$:=$	$\top$	top
$\llbracket \perp \rrbracket_\alpha$	$:=$	$\perp$	bottom
$\llbracket \varphi \vee \psi \rrbracket_\alpha$	$:=$	$\llbracket \varphi \rrbracket_\alpha \cup \llbracket \psi \rrbracket_\alpha$	join
$\llbracket \varphi \wedge \psi \rrbracket_\alpha$	$:=$	$\llbracket \varphi \rrbracket_\alpha \cap \llbracket \psi \rrbracket_\alpha$	meet
$\llbracket \neg \varphi \rrbracket_\alpha$	$:=$	$\sim \llbracket \varphi \rrbracket_\alpha$	complement

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On the assumption $\mathcal{A}$ is a **complete** Boolean algebra, we let:

$$\llbracket \exists p \varphi(p) \rrbracket_\alpha := \bigcup_{a \in A} \llbracket \varphi \rrbracket_{\alpha[a/p]}$$

Existential quantification is akin to generalized disjunction.

Consistency

Complete Boolean algebras support Boolean-valued models of QPL interpreting existential quantification in terms of join.

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Ideological Expansion ...

We now expand $\mathcal{L}$ with a propositional operator Q . In line with [Kaplan, 1995], we interpret $Q\varphi$ as: *φ is uniquely queried*.

We let Q be governed by two simple axioms:

$$\begin{array}{ll} \forall p (Qp \neq \perp) & \text{Regularity} \\ \forall p \forall q (Qp \wedge Qq \rightarrow p = q) & \text{Relative Atomicity} \end{array}$$

In words:

- Whatever p may be, it is not absurd to uniquely query p .
- No two distinct propositions are uniquely queried.

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Generalizing the Algebraic Semantics

Expand a Boolean algebra $(A, \leq)$ with a function f and set:

$$\llbracket Q\varphi \rrbracket_\alpha := f(\llbracket \varphi \rrbracket_\alpha)$$

For $(A, \leq, f)$ to model QPL + Regularity + Relative Atomicity, it must satisfy two conditions:

Reg. $\forall x \in A, f(x) \neq \perp$.

RAt. $\forall x \in A \forall y \in A (x \neq y \rightarrow f(x) \cap f(y) = \perp)$

Call $(A, \leq, f)$ a *well-behaved* iff it satisfies both constraints.

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Incompleteness

No well-behaved expansion $(A, \leq, f)$ of a Boolean algebra $(A, \leq)$ is complete.

If $(A, \leq, f)$ is a well-behaved expansion, then the range of f on A forms an *anti-chain*, i.e., a family of pairwise disjoint (given RAt) non-zero (given Reg) elements, of the same cardinality as A .

No two distinct subsets of an anti-chain can share a join in the algebra. Since the anti-chain has the cardinality of A , not all of its subsets have a join in $(A, \leq, f)$.

This may make it sound as a cardinality issue, but the problem is more general.

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... and Paradox

If A is infinite, then there are no more $\mathcal{L}_Q$ -definable subsets of A than elements of A . Yet ...

Non-closure under definable join

No well-behaved expansion $(A, \leq, f)$ of a Boolean algebra $(A, \leq)$ is closed under $\mathcal{L}_Q$ -definable join.

Consider the condition:

$$Qp \wedge \neg p$$

Nothing can be a join of instances of the form:

$$\bigcup_{a \in A} (f(a) \cap \sim a).$$

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Assume for reductio that some r is $\bigcup_{a \in A} (f(a) \cap \sim a)$.

i. $f(r) \cap \sim r \leq r$ and $f(r) \cap \sim r \leq \sim r$.

ii. $f(r) \cap \sim r = \perp$, and

$$r = \perp \cup \bigcup_{a \neq r} (f(a) \cap \sim a)$$

iii. from ii., we have $f(r) \leq r$ and $f(r) = f(r) \cap r$. So:

$$f(r) = f(r) \cap \bigcup_{a \neq r} (f(a) \cap \sim a).$$

iv. By **RAt**, $f(r) \cap f(a) = \perp$ for all $a \neq r$ and:

$$f(r) = f(r) \cap \bigcup_{a \neq r} (f(a) \cap \sim a) = \perp.$$

This contradicts **Reg** above.

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Boolean unsatisfiability

No complete Boolean algebra supports a Boolean-valued model of QPL + Regularity + Relative Atomicity

This is not a limitation of the algebraic semantics, but rather a symptom of a more serious issue:

$$\begin{aligned} \bigcup_{a \in A} f(a) \cap \sim a &= \bigcup_{a \in A} \llbracket Qp \wedge \neg p \rrbracket_{[a/p]} \\ &= \llbracket \exists p (Qp \wedge \neg p) \rrbracket \end{aligned}$$

But $\exists p (Qp \wedge \neg p)$ plays a crucial role in Prior's Theorem ...

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Prior's Theorem

Prior's Theorem

$$\vdash Q \exists p (Qp \wedge \neg p) \rightarrow \exists p (Qp \wedge p) \wedge \exists p (Qp \wedge \neg p)$$

Let λ be $\exists p (Qp \wedge \neg p)$. Given classical quantificational logic, $\vdash (Q\lambda \wedge \neg\lambda) \rightarrow \lambda$, whence $\vdash Q\lambda \rightarrow \lambda$. It follows that $\vdash Q\lambda \rightarrow (Q\lambda \wedge \lambda)$ and $\vdash Q\lambda \rightarrow \exists p (Qp \wedge p)$. On the other hand, $\vdash \lambda \rightarrow \exists p (Qp \wedge \neg p)$.

The existence of such a proposition as $\exists p (Qp \wedge \neg p)$ is inconsistent with Regularity and Relative Atomicity.

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From Prior to Kaplan

Inconsistency

QPL + Regularity + Relative Atomicity is inconsistent.

By Regularity, $Q\lambda \neq \perp$. By Prior's theorem, we have both $\exists p (Qp \wedge p) \neq \perp$ and $\exists p (Qp \wedge \neg p) \neq \perp$. But if μ and ν such that $(Q\mu \wedge \mu)$ and $(Q\nu \wedge \neg\nu)$, then $Q\mu \wedge Q\nu \neq \perp$ and $\mu \neq \nu$, which conflicts with Relative Atomicity.

This is closely related to Kaplan's paradox when we interpret $Q\varphi$ as: φ is *possibly* uniquely Q .

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Two Responses: Expansion and Refinement

How to Respond

There are two main lines of response:

- **follow the argument where it leads**
We learn that no propositional attitude can verify Reg and RelAt on pain of contradiction. One must fail for *being uniquely queried*, e.g., some proposition is such that it is simply absurd to uniquely query it. [Bacon et al., 2016]
- **attempt to block the argument**
Revise classical quantificational logic in order to block Prior's theorem, e.g., restrict universal instantiation. [Kaplan, 1995]

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How to Model Ramification

One response, in line with [Kaplan, 1995], is to ramify:

- Propositions of level 0 concern *extensional matters*, e.g., distributions of earth, air, fire, and water.
- Propositions of level 1 concern both extensional matters and patterns of application of Q to propositions of level 0.
- Propositions of level 2 include propositions of level 1 and patterns of application of Q to level 1.

Here is how [Bacon et al., 2016] implement the proposal:

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Finite Possibilities

Given a set W_0 of proto-worlds, finite possibilities are sequences:

$$\begin{array}{ll} S_0 = W_0 & \mathcal{I}_0 = \mathcal{P}(S_0) \\ S_1 = W_0 \times \mathcal{I}_0 & \mathcal{I}_1 \subseteq \mathcal{P}(S_1) \end{array}$$

$$S_n = W_0 \times \mathcal{I}_0 \times \cdots \times \mathcal{I}_{n-1} \quad \mathcal{I}_n \subseteq \mathcal{P}(S_n)$$

S_n consists of stage n possibilities $(a, E_1, \dots, E_{n-1})$, which:

- specify a proto-world a
- record which stage m intensions, for $m < n$, are Q . $E_m \subseteq \mathcal{I}_m$ is the set of stage m intensions that are Q .

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Finite Intensions

As a first approximation, stage n intensions are *informative* sets of stage n possibilities.

Not all sets of stage n possibilities carry new information not encoded by earlier sets, e.g., $\mathcal{S}_1$ carries no information beyond $\mathcal{S}_0$.

Stage n intensions are sets of stage n possibilities, which carry *new* information not encoded by earlier sets of possibilities.

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Worlds are Complete Possibilities

We now let a *world* w be an ω -sequence of the form:

$$(a, E_0, \dots, E_n, \dots),$$

where a is a proto-world and E_n is a set of stage n intensions.

Call w a *completion* of a stage n possibility s iff $w \upharpoonright n = s$.

W is the set of completions of finite stage possibilities.

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Propositions are Complete Intensions

We let the *completion* $\hat{I}$ of a stage n possibility X be the set of world completions of stage n possibilities in X .

$$\hat{X} = \{w : w \upharpoonright n \in X\}$$

intension for some $m \leq n$.

If $\hat{I}$ is the completion of a stage n intension I , then:

- $\hat{I}$ agrees with I up to stage n .
- $\hat{I}$ is noncommittal after stage n .

Completions are closed under Boolean operations.

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A Tower of Boolean Algebras

We have a tower of Boolean algebras

$$(A_n, \cap_n, \cup_n, \sim_n, \emptyset, W)$$

where A_n is the set of completions of stage $\leq n$ intensions, and for each $n \in \omega$, $A_n \subset A_{n+1}$.

We have ramified propositions into cumulative levels: propositions of level n are completions of stage m intensions for $m \leq n$.

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A Semantics for Ramification

An assignment α maps propositional variables p_n of level n to completions of stage $\leq n$ intensions.

$\llbracket p_n \rrbracket_\alpha^n$	$:= \alpha(p_n)$	
$\llbracket \top \rrbracket_\alpha^n$	$:= W$	top
$\llbracket \perp \rrbracket_\alpha^n$	$:= \emptyset$	bottom
$\llbracket \varphi \vee \psi \rrbracket_\alpha^n$	$:= \llbracket \varphi \rrbracket_\alpha^n \cup \llbracket \psi \rrbracket_\alpha^n$	join
$\llbracket \varphi \wedge \psi \rrbracket_\alpha^n$	$:= \llbracket \varphi \rrbracket_\alpha^n \cap \llbracket \psi \rrbracket_\alpha^n$	meet
$\llbracket \neg \varphi \rrbracket_\alpha^n$	$:= \sim_n \llbracket \varphi \rrbracket_\alpha^n$	complement

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A Semantic Clause for Q

We expand the Boolean algebras with a function f :

$$f(\hat{l}) = \{w : l \in E_n\}$$

f maps a completion $\hat{l}$ of a stage n intension l to a set of worlds in which E_n contains the stage n intension l . We now set:

$$\llbracket Q\varphi \rrbracket_\alpha^n := f(\llbracket \varphi \rrbracket_\alpha^n)$$

Two observations:

- Regularity is satisfied, since $f(\hat{l}) \neq \emptyset$ for each completion $\hat{l}$.
- Relative Atomicity requires a restriction of W to worlds at which *all but at most one* E_n are empty and E_n is a singleton.

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Ramified Prior propositions

There is a Prior proposition

$$\exists p_n(Qp_n \wedge \neg p_n) \quad (\lambda_n)$$

for each level of the hierarchy. Note that each Prior proposition λ_n quantifies over propositions of level n .

What exactly is the level of the Prior proposition λ_n ? It is **not** n .

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How to Compute λ_n

If $\alpha(p_n)$ is a completion $\hat{l}$ of some stage $m \leq n$ intension, then:

$$\begin{aligned} \llbracket Qp_n \wedge \neg p_n \rrbracket_\alpha^n &= \llbracket Qp_n \rrbracket_\alpha^n \cap \llbracket \neg p_n \rrbracket_\alpha^n \\ &= f(\hat{l}) \cap_n \sim_n \hat{l} \\ &= \{w : l \in E_m\} \cap \{w : w \upharpoonright m \notin l\} \\ &= \{w : l \in E_m \wedge w \upharpoonright m \notin l\} \end{aligned}$$

Existential quantification is akin to generalized disjunction:

$$\llbracket \exists p_n(Qp_n \wedge \neg p_n) \rrbracket^n = \bigcup_{\hat{l} \in A_n} \{w : l \in E_m \wedge w \upharpoonright m \notin l\}$$

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Observation

$\llbracket \lambda_n \rrbracket^n$ is **not** a completion of a stage $m \leq n$ intension.

Existential generalization fails in the model. Since λ_n is itself **not** a proposition of level n , we are **not** entitled to:

$$(Q\lambda_n \wedge \lambda_n) \rightarrow \exists p_n (Qp_n \wedge \neg p_n).$$

The proof of Prior's theorem is blocked.

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Suppose, for a reductio, that $\llbracket \lambda_n \rrbracket^n$ is the completion $\hat{J}$ of some stage $m \leq n$ intension. $\llbracket \lambda_n \rrbracket^n \neq \emptyset$, e.g., the world completion of $(a, \{\{a\}\})$ is in $\llbracket \lambda_n \rrbracket^n$. Choose a world w for which

$$w \restriction m \in J \quad \text{and} \quad J \in E_m.$$

Is w a member of $\llbracket \lambda_n \rrbracket^n$?

$$\begin{aligned} w \in \llbracket \lambda_n \rrbracket^n &\Leftrightarrow w \in \hat{J} \\ &\Leftrightarrow w \restriction m \in J \\ &\Leftrightarrow w \restriction m \in J \vee J \notin E_m \\ &\Leftrightarrow w \notin \llbracket \lambda_n \rrbracket^n \end{aligned}$$

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Summary

Some observations:

- The proof of Prior's Theorem is blocked when we exclude λ_n from the range of the existential quantifier $\exists p_n$.
- None of the algebras in the tower

$$(A_n, \cap_n, \cup_n, \sim_n, \emptyset, W, f)$$

is closed under definable join.

- Regularity is satisfied.
- Relative Atomicity requires a further restriction on W .

We now look at a different strategy ...

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Refinement

We now keep the domain constant, and look at successive refinements of the order. That is, we consider a series of pre-ordered sets of the form:

$$\left(\bigcup_{n \in \omega} A_n, \sqsubseteq_n \right)$$

For each stage n , define a preorder $\sqsubseteq_n$ on $\bigcup_{n \in \omega} A_n$:

$$\hat{I} \sqsubseteq_n \hat{J} \quad \text{iff} \quad \hat{I} \restriction n \subseteq \hat{J} \restriction n$$

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Indiscernibility

The preorder determines an equivalence relation $\equiv_n$:

$$\widehat{I} \equiv_n \widehat{J} \quad \text{iff} \quad \widehat{I}|_n = \widehat{J}|_n$$

If $\widehat{I} \equiv_n \widehat{J}$, then they are *indiscernible* at stage n .

As we ascent into the hierarchy, we refine our powers of discrimination. At stage n , we remain blind to distinctions that only become apparent at later stages.

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The Quotient Algebra

We have a Boolean algebra $(\bigcup_{n \in \omega} A_n / \equiv_n, \leq_n)$ for each stage n .

$$\begin{aligned} [\widehat{I}]_n \cup_n [\widehat{J}]_n &= [\widehat{I} \cup \widehat{J}]_n && \text{join} \\ [\widehat{I}]_n \cap_n [\widehat{J}]_n &= [\widehat{I} \cap \widehat{J}]_n && \text{meet} \\ \sim_n [\widehat{I}]_n &= [\sim \widehat{I}]_n && \text{complement} \\ \top_n &= [W]_n \\ \perp_n &= [\emptyset]_n \end{aligned}$$

We may choose a canonical representative for each equivalence class

$$c([\widehat{I}]_n) = \widehat{I}|_n.$$

$\widehat{I}|_n$ is a completion of some stage $m \leq n$ intension.

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Individuation

A completion $\widehat{I}$ is **individuated at stage n** if it is identical to its canonical representative at that stage:

$$\widehat{I} = c([\widehat{I}]_n).$$

This is the case when $\widehat{I}$ is a completion of a stage n intension.

A completion $\widehat{I}$ is **individuated by stage n** iff $\widehat{I}$ is individuated at some earlier stage $m \leq n$.

Some observations:

- Both W and $\emptyset$ have been individuated at stage 0.
- Once $\widehat{I}$ is individuated at stage n , it remains individuated by a later stage.

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Refinement Semantics

Given the pre-order:

$$(\bigcup_{n \in \omega} A_n, \sqsubseteq_n),$$

we let α map propositional variables into $\bigcup_{n \in \omega} A_n$:

$$\begin{aligned} \llbracket p \rrbracket_\alpha^n &:= \alpha(p) \\ \llbracket \neg \varphi \rrbracket_\alpha^n &:= \sim_n c(\llbracket \varphi \rrbracket_n) && \text{complement} \\ \llbracket \varphi \vee \psi \rrbracket_\alpha^n &:= c(\llbracket \varphi \rrbracket_n) \cup_n c(\llbracket \psi \rrbracket_n) && \text{join} \\ \llbracket \varphi \wedge \psi \rrbracket_\alpha^n &:= c(\llbracket \varphi \rrbracket_n) \cap_n c(\llbracket \psi \rrbracket_n) && \text{meet} \end{aligned}$$

Variables range over the constant domain, but the Boolean operations are sensitive only to the discernible behavior of a proposition.

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Two More Semantic Clauses

Existential quantification is still interpreted in terms of join.

$$\llbracket \exists p \varphi \rrbracket_\alpha^n := \bigcup_{\hat{I} \in \bigcup_n A_n} c(\llbracket \varphi \rrbracket_\alpha^n[\hat{I}/p])$$

We now expand the pre-orders with the function f defined above:

$$(\bigcup_{n \in \omega} A_n, \sqsubseteq_n, f)$$

The semantic clause for Q is as follows:

$$\llbracket Q\varphi \rrbracket_\alpha^n := f(c(\llbracket \varphi \rrbracket_n))$$

Notice that $\llbracket Q\varphi \rrbracket_\alpha^n$ is only sensitive to the discernible behavior of $\llbracket \varphi \rrbracket_\alpha^n$ at the given stage.

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The Prior proposition

There is now one Prior proposition $\exists p(Qp \wedge \neg p)$ abbreviated: λ .

Observation

$\llbracket \lambda \rrbracket^n$ is **not** a completion of a stage $m \leq n$ intension.

λ has not yet been individuated at stage n . This matters because:

Observation

If $\hat{I}$ has **not** been individuated at stage n , then $f(\hat{I} \upharpoonright n) \equiv_n \top$

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This is because:

$$f(\hat{I} \upharpoonright n) = \{w : I \upharpoonright n \in E_n\}$$

Any stage n possibility $(a, E_0, \dots, E_{n-1})$ may be extended into one in which $I \upharpoonright n \in E_n$, e.g., $(a, E_0, \dots, E_{n-1}, \{I \upharpoonright n\})$. Therefore:

$$\begin{aligned} f(\hat{I} \upharpoonright n) \upharpoonright n &= \{w : w \upharpoonright n \in \mathcal{S}_n\} \\ &= W \upharpoonright n \end{aligned}$$

Thus:

$$f(\hat{I} \upharpoonright n) \equiv_n \top$$

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Since neither $\llbracket \lambda \rrbracket^n$ nor $\llbracket \neg \lambda \rrbracket^n$ are individuated by stage n :

Observation

$$\llbracket Q\lambda \rrbracket^n \equiv_n \top \equiv_n \llbracket Q\neg\lambda \rrbracket^n$$

Two observations:

- $Q\lambda$ and $Q\neg\lambda$ are, relative to all finite stages, indiscernible form $\top$. This will help us vindicate Prior's Theorem.
- If $\hat{I}$ is **not** yet individuated at stage n , then we will have $\llbracket Qp \wedge Q\neg p \rrbracket_\alpha^n[\hat{I}/p] \equiv_n \top$ even if we restrict the domain of worlds in order to accommodate RA_t.

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Prior Revisited

Recall Prior's Theorem:

$$Q\exists p(Qp \wedge \neg p) \rightarrow \exists p(Qp \wedge p) \wedge \exists p(Qp \wedge \neg p)$$

The antecedent $Q\lambda$ is n -indiscernible from $\top$ at all stages of refinement as is $Q\neg\lambda$. Since $\lambda \vee \neg\lambda$ is $\top$ relative to all stages of refinement, both $\exists p(Qp \wedge p)$ and $\exists p(Qp \wedge \neg p)$ become indiscernible from $\top$ at every finite stage of refinement.

Both λ and $\neg\lambda$ provide witnesses to the existential generalizations in the consequent, but the reason is that neither $Q\lambda$ nor $Q\neg\lambda$ have come into focus at stage n .

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Summary

- The pre-ordered sets

$$(\bigcup_{n \in \omega} A_n, \sqsubseteq_n)$$

are **closed under definable join**. When we compute the join of a definable set, we only input the canonical representatives at stage n , which results in a completion of a stage $m \leq n + 1$ intension.

The join in question may not yet individuated at stage n . That is exactly the situation with $\llbracket \lambda \rrbracket^n$, which is never individuated at stage n .

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- Regularity is satisfied: $f(\hat{I}) \neq \emptyset$ for all $\hat{I}$ in $\bigcup_{n \in \omega} A_n$.
- Relative Atomicity collapses for propositions newly individuated at a given finite stage, but it is still satisfied by propositions individuated at strictly earlier stages when we restrict the domain of worlds to contain only worlds at which exactly one finite stage intension figures in some E_n .

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Conclusion

We distinguish two approaches to the risks of ideological expansion.

A strategy of **ontological expansion** involves ramification and requires a restriction on existential generalization. Since λ_n lies outside the range of $\exists p_n$, we are not entitled to the application of existential generalization required for the proof of the theorem.

A strategy of **refinement** preserves classical logic and frames Prior's Theorem as a byproduct of unavoidable discriminatory limitations on our part. There is a fixed domain of propositions, but our ability to discriminate among them evolves through a succession of preorders, which correspond to ever more fine-grained relations of entailment on that domain.

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